

NORTHERN ELECTRICAL TESTING COMPANY (NPCETC)

Hanoi, Vietnam | Tel.: +84 24 3875 9361 | npcetc.vn

Ha Noi, July 22, 2026

To: Ph.D. Selection Committee
Doctoral Program in Electrical, Electronics and Computer Engineering
College of Engineering and Computer Science (CECS)
University of Michigan–Dearborn, MI 48128, USA

Dear Admissions Committee,

We are writing to offer our strongest recommendation for Xuan Thanh Trinh's application to your Ph.D. program. We had the privilege of mentoring Thanh during his technical internship at the Relay Testing Department of the Northern Power Electrical Testing Company (NPCETC) in the summer of 2025. While one of us (Van Dinh Thanh) directly supervised his daily engineering analyses, the other (Trinh Xuan Long) oversaw his overall internship performance and deliverables. Based on his outstanding performance, we are confident that he is thoroughly prepared for the rigors of doctoral research.

What distinguished Thanh from other interns was his clear ability to decode complex physical grid systems purely through analytical observation, proving he possesses a highly robust technical foundation. Because strict industrial safety protocols rightfully restricted him from directly operating high-voltage equipment, he focused his technical acumen on dissecting dense industrial schematics. Specifically, he analyzed the complex secondary circuits and protection protocols of the 110kV Tam Dao substation. Furthermore, he demonstrated a strong capacity to logically trace these physical mechanisms in the field, accurately understanding how hardware interlock constraints directly dictate the configuration of numerical protective relays. This capability to mentally simulate and analyze physical systems through blueprints and field observations proves he has the analytical foundation needed for rigorous, simulation-driven power system modeling.

Beyond his analytical capabilities, Thanh exhibited a strong commitment to continuous learning and a high degree of professional maturity within our demanding corporate environment. He proactively shadowed our senior field engineers, displaying active technical curiosity by asking targeted questions that bridged his academic coursework with our practical, safety-critical field constraints. Additionally, he meticulously synthesized these field interactions and technical observations into a highly structured internship report, demonstrating a level of technical communication and clarity rarely seen in undergraduate engineering students. His coachability, combined with his ability to absorb and articulate complex industrial concepts, guarantees that he will be a highly collaborative and reliable addition to any academic research laboratory.

In summary, Thanh uniquely combines rigorous analytical skills with outstanding professional maturity. We are fully confident that he will be a strong addition to your academic research group. Therefore, we give him our highest recommendation for admission without reservation. Should you require any further information, please feel free to contact us.

Sincerely,

Trinh Xuan Long
Head of Human Resources
Northern Electrical Testing Company
Email: longtx@npcetc.vn

Van Dinh Thanh
Direct Supervisor (Instructor)
Northern Electrical Testing Company
Email: thanhvd@npcetc.vn